Semi-Supervised Teeth and Root Pulp Canal Segmentation in CBCT Scans via nnU-Net Self-Training

Independent participation — submitted outside the official evaluation window

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Abstract

This report describes an independent submission to Task 1 of the MICCAI STSR 2026 Challenge: semi-supervised segmentation of permanent teeth and their root pulp canals in 3D CBCT scans, across 65 label classes. The approach is a two-stage self-training pipeline built on nnU-Net v2. A fully supervised 3d_fullres model was first trained on the 30 annotated scans and used to generate pseudo-labels for the 299 unannotated scans; a 5-fold ensemble was then fine-tuned for 250 epochs per fold on the combined 329-case dataset, initialized from the supervised checkpoint. Inference runs each fold as an isolated process and combines the five label maps by majority vote, a memory-efficient approximation of softmax averaging.

Performance was assessed by fold-aware cross-validation on all 30 annotated cases: each case was predicted exclusively by the single fold that had excluded it from training, avoiding the optimistic bias of evaluating with folds that had seen the case. Mean Dice over ground-truth-present classes was 0.619 ± 0.128 (median 0.632; range 0.378–0.845; 30/30 cases completed without failure). Separating the label groups shows a consistent gap: 0.725 ± 0.158 for tooth classes versus 0.473 ± 0.163 for root pulp canal classes. Qualitative review indicates that the dominant residual error is tooth identity assignment (ADA numbering) rather than boundary delineation, and that the weakest cases coincide with severe metallic streak artifacts — consistent with the fact that no metal artifact reduction was implemented. Predictions for all 20 official validation cases were produced and packaged in the required format, but could not be submitted because both submission windows had closed.

1. Introduction

Segmentation of teeth and root pulp canals in CBCT supports endodontic planning and digital dental modeling, but manual delineation is slow and expertise-dependent, and metallic restorations introduce streak artifacts that degrade image quality. The MICCAI STSR 2026 Task 1 challenge frames this as a semi-supervised problem: a small annotated set alongside a much larger unannotated pool, evaluated on a held-out validation set [1,2,3,4].

This report documents an independent submission. The author is a dentist (DDS, Universidade de São Paulo) participating in an AI-focused study group; this was a first end-to-end machine learning pipeline applied to clinical imaging data. Registration occurred on the final day of the registration window, leaving minimal time to iterate: an initial custom training pipeline was abandoned mid-course after a silent optimizer configuration fault, and by the time a working nnU-Net-based pipeline had been validated, both the Validation-phase and the Docker-phase submission deadlines had passed. No official score was therefore obtained. The purpose of this report is to document that the dataset and the evaluation protocol were used correctly and completely, to report honestly measured results, and to request feedback ahead of STSR 2027.

2. Data and Task

The Task 1 dataset comprises 30 CBCT volumes with expert annotations, 299 CBCT volumes without annotations, and 20 CBCT volumes forming the official validation set (ground truth not released to participants). The label space contains 65 classes: background, 32 permanent-tooth classes following ADA numbering (including third molars), and 32 corresponding root pulp canal classes, where class N denotes a tooth (N = 1…32) and class N+32 denotes the root pulp canal of that same tooth. Median voxel spacing in the released data is 0.30 mm isotropic, with a median volume of 524 × 400 × 524 voxels.

3. Method

3.1 Pipeline overview

The final pipeline is two-stage self-training on nnU-Net v2 [5]: (i) a supervised model trained on the 30 annotated cases generates pseudo-labels for the 299 unannotated cases; (ii) a 5-fold ensemble is fine-tuned on the union of real and pseudo-labeled data, initialized from the stage-1 checkpoint. This follows the general strategy that nnU-Net-based entries have used in previous editions of this challenge series [2,3,4].

3.2 Stage 1 — supervised baseline

A 3d_fullres model was trained on the 30 annotated cases using nnU-Net's standard auto-configuration, with no architectural modification. This checkpoint served both to generate pseudo-labels and to initialize the final ensemble.

3.3 Pseudo-label generation

Pseudo-labels for the 299 unannotated cases were produced by invoking the nnU-Net Python inference API directly (predict_single_npy_array), one case per process, using the stage-1 checkpoint with a single fold and without test-time augmentation, retaining only the argmax label map. No voxel-level or case-level confidence filtering was applied (Section 7). The command-line predictor was evaluated first, but its internal multiprocessing export stage stalled reproducibly in this environment across all configurations attempted, which motivated the single-process API call.

3.4 Semi-supervised dataset assembly

The 30 annotated and 299 pseudo-labeled cases were merged into a single 329-case nnU-Net dataset using filesystem hard links to avoid duplicating volumetric data on a size-constrained volume. Because nnU-Net derives a dataset-specific plan, the plan generated for the merged dataset was incompatible with the stage-1 checkpoint; the plan from the 30-case dataset was therefore transferred to the merged dataset (nnUNetv2_move_plans_between_datasets) and all 329 cases were re-preprocessed under it, allowing the stage-1 weights to be reused without architectural mismatch.

3.5 Ensemble fine-tuning

Five folds were fine-tuned independently on the 329-case dataset, each initialized from the stage-1 checkpoint. The configuration below is taken from the plans file stored alongside the trained model, and therefore reflects what training actually used.

Parameter	Value
Framework / configuration	nnU-Net v2, 3d_fullres
Architecture	PlainConvUNet, 6 stages, features 32/64/128/256/320/320
Convolution kernels	3 × 3 × 3 at all stages; 2 convolutions per stage (encoder and decoder)
Downsampling strides	1,1,1 → 2,2,2 (×4) → 2,1,2
Normalization / activation	InstanceNorm3d (affine) / LeakyReLU; no dropout
Patch size	128 × 96 × 160
Batch size	2 (batch Dice enabled)
Target spacing	0.30 × 0.30 × 0.30 mm (isotropic)
Median size at target spacing	480 × 333 × 498 voxels
Intensity normalization	CTNormalization (foreground mean 1748.4, SD 762.5; 0.5–99.5 percentile 144–3945)
Resampling	cubic (order 3) for images; linear (order 1) for labels and probabilities
Training length	250 epochs per fold (nnUNetTrainer_250epochs)
Initialization	stage-1 supervised checkpoint (pretrained weights)
Loss, optimizer, LR schedule, augmentation	nnU-Net v2 defaults, unmodified (Dice + cross-entropy with deep supervision; SGD with Nesterov momentum; polynomial LR decay)

Table 1. Training configuration, read from the plans file stored with the trained model.

3.6 Inference and ensembling

At inference, each of the five folds is executed as an isolated operating-system subprocess for a given case, and the five resulting label maps are combined by majority vote per voxel. This approximates nnU-Net's standard ensembling (averaging softmax probabilities before argmax) while avoiding the need to hold 65-class probability volumes for five folds in memory simultaneously, which exceeded available memory in earlier attempts. The approximation is a deliberate accuracy-for-memory trade-off and is discussed in Section 7.

4. Experimental Setup

All experiments ran on cloud GPU instances with a persistent network-attached volume, allowing the GPU type to change mid-project without loss of data or checkpoints. Three configurations were used:

GPU	VRAM	Container RAM	Role
RTX 4090	24 GB	~77 GB	Stage-1 training, pseudo-label generation, preprocessing, 5-fold ensemble training
RTX PRO 6000 (Blackwell)	~96 GB	~175 GB	Adopted after out-of-memory failures when reconstructing full-resolution 65-class probability volumes; final 20-case inference
A100 SXM	80 GB	~116 GB	Cross-validation experiments and submission packaging

Table 2. Hardware used across the project.

Two engineering practices proved necessary. First, each unit of work (one case, one fold) was executed as an isolated subprocess rather than within a shared multiprocessing pool, because pool-based parallelism repeatedly either stalled or propagated a single worker's out-of-memory failure to the entire batch. Second, resource ceilings were read directly from the container's cgroup interface rather than from host-level utilities, which report the physical machine's totals rather than the container's allocation and led to incorrect parallelism assumptions on more than one occasion. Even on the 175 GB configuration, running two folds concurrently for a single case caused out-of-memory failures on the largest volumes, so folds were run strictly sequentially.

5. Evaluation Protocol

Because all 30 annotated cases were included in the 329-case training set, evaluating them with the full 5-fold ensemble would be optimistically biased: for any given case, four of the five folds had seen it during training. To obtain an unbiased estimate, a fold-aware cross-validation was performed — each annotated case was predicted exclusively by the single fold whose training split excluded it, as determined from the dataset's stored split file.

The reported metric is the Dice coefficient computed per class and averaged over the classes present in that case's ground truth. Two properties of this definition should be noted. It does not penalize predicted classes that are absent from the ground truth (false-positive classes), and it weights every present class equally regardless of volume, so a small pulp canal counts as much as a molar crown. All results in Section 6 were computed by a single script under this one definition; earlier ad-hoc measurements taken during development used a different class-selection rule and are therefore not reported here, to avoid mixing incompatible numbers.

This protocol measures a single model, not the five-fold ensemble that was actually submitted. Since ensembling generally improves on any individual member, these figures should be read as a conservative lower bound on the submitted configuration's performance. The submitted ensemble itself cannot be scored, as ground truth for the 20 official validation cases is not public.

6. Results

6.1 Cross-validated accuracy (n = 30)

Label group	Mean	SD	Median	Min	Max
All classes	0.619	0.128	0.632	0.378	0.845
Tooth classes (1–32)	0.725	0.158	0.758	0.326	0.920
Pulp canal classes (33–64)	0.473	0.163	0.456	0.211	0.790

Table 3. Fold-aware cross-validated Dice across all 30 annotated cases.

All 30 cases completed without failure. The distribution is broad but not bimodal: 10 cases scored ≥ 0.70, 14 between 0.50 and 0.70, and 6 below 0.50. The gap of 0.252 Dice between tooth classes and their corresponding pulp canal classes is the most consistent finding in the data, present in 26 of the 30 cases, and is expected given that pulp canals are substantially thinner structures.

6.2 Dominant failure mode: identity assignment rather than delineation

Visual review of the weakest cases indicates that predicted masks generally follow tooth boundaries correctly but are assigned a different class index than the ground truth — typically that of an adjacent tooth. Because Dice is computed per class, a one-position shift in ADA numbering drives the Dice of both the affected classes to zero even though the underlying segmentation is anatomically plausible. This is corroborated quantitatively by four cases in which pulp canal Dice exceeds tooth Dice, inverting the pattern seen in the other 26; the two weakest cases overall (0.378 and 0.398) both show this inversion, with tooth Dice collapsing to 0.326 and 0.342 while their pulp scores remain near the cohort median.

This is consistent with the published final Docker rankings for this challenge, where the leading team reached 0.870 image-level Dice but only 0.541 instance affinity, and no team exceeded 0.60 on that metric. Correct assignment of tooth identity, rather than boundary accuracy, appears to be the discriminating difficulty of this task across participants.

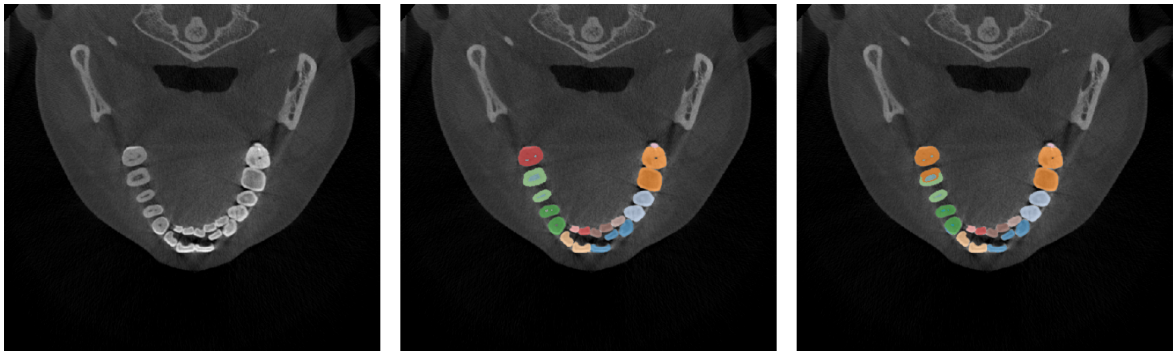


Figure 1. Representative case (case_0005). Left: axial CT slice. Centre: ground truth. Right: 5-fold majority-vote prediction. Colour encodes class identity. Predicted masks follow tooth boundaries closely, but several teeth are assigned a different class index than the reference, illustrating the identity-assignment error mode. Fold-aware cross-validated Dice for this case: 0.526.

6.3 Metallic artifacts

The lowest-scoring cases coincide with severe metallic streak artifacts. In the case shown in Figure 2, pronounced star-burst streaking crosses the entire mandible; the predicted masks track visible tooth structures but assign them identities that diverge from the reference. Since no metal artifact reduction component was implemented — although it forms part of the STSR 2026 Task 1 scope — this behaviour is an expected consequence of the pipeline's design rather than an unexplained anomaly.

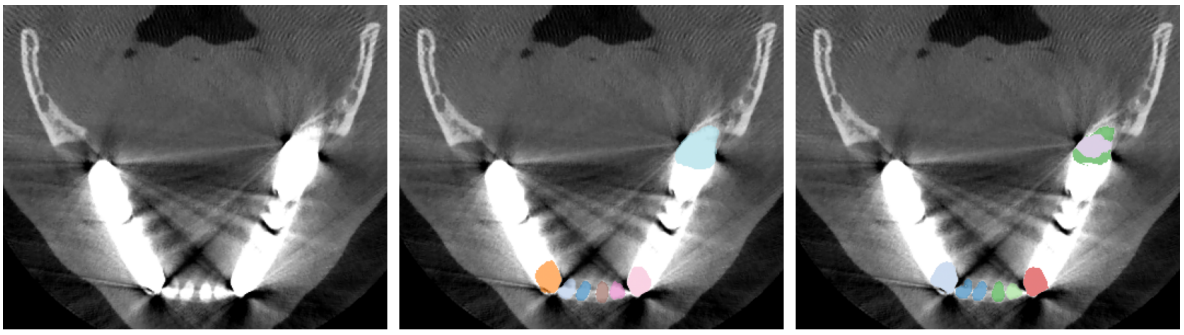


Figure 2. Case with severe metallic streak artifacts (case_0023). Left: axial CT slice. Centre: ground truth. Right: 5-fold majority-vote prediction. This scan carries annotations for only 14 of the 65 classes, so per-case averages over present classes are computed from a small set. Fold-aware cross-validated Dice for this case: 0.501.

6.4 Submission package and computational cost

Predictions were generated for all 20 official validation cases using the 5-fold majority-vote ensemble, and packaged according to the Validation-phase specification: 20 files named STS26_Validation_xxxx.nii.gz (with the _with-artifacts suffix where applicable), archived without subfolders. The package was verified against the published specification but could not be submitted, as the Validation-phase window had closed. No Docker container was built for the ranked Docker phase; containerizing the full inference pipeline was a separate engineering effort not undertaken in the time available.

End-to-end inference cost for the submitted configuration was approximately 20 minutes per case on the RTX PRO 6000 configuration, dominated by five sequential fold passes and by CPU-bound resampling of the 65-class probability volume back to native resolution. Peak GPU memory during single-fold inference was approximately 44 GB.

7. Limitations

- No confidence-based filtering of pseudo-labels. A voxel-level confidence filter was implemented but never connected to the pipeline that produced the final training set; all 299 pseudo-labeled cases entered training as raw argmax label maps, propagating stage-1 errors, including systematic tooth-numbering errors, into the ensemble.
- No metal artifact reduction was implemented, despite being part of the Task 1 scope; the weakest cases in cross-validation are artifact-heavy scans.
- Ensembling by majority vote rather than softmax averaging is an approximation adopted for memory reasons and is expected to cost some accuracy relative to nnU-Net's standard strategy.
- The reported metric does not penalize predicted classes absent from the ground truth, and weights all present classes equally regardless of volume. Complementary metrics used by the challenge — NSD, mIoU, and instance affinity — were not computed, so these results are not directly comparable to the official leaderboard.
- Cross-validation measures single-fold models; the submitted 5-fold ensemble was never evaluated against ground truth, since none is public for the official validation set.
- Inference cost (~20 minutes per case) is far above what a time-constrained Docker evaluation would require.
- Registration on the final day of the registration window left no margin to meet either submission deadline.

8. Conclusion and Future Work

A complete semi-supervised segmentation pipeline was implemented and evaluated: supervised pretraining, pseudo-label self-training over 299 unannotated scans, 5-fold ensemble fine-tuning, fold-aware cross-validation on all 30 annotated cases, and inference on all 20 official validation cases in the required submission format. Cross-validated accuracy was 0.619 ± 0.128 mean Dice, with tooth classes at 0.725 and pulp canal classes at 0.473, and no case-level failures.

The analysis points to a clear priority for future work: the limiting factor is tooth identity assignment, not boundary delineation. Concretely, this suggests adding confidence filtering to pseudo-labels so that stage-1 numbering errors are not propagated, introducing an explicit instance-level or anatomically constrained numbering step downstream of segmentation, and incorporating metal artifact reduction. A separate engineering track concerns inference cost: keeping fold models

resident on the GPU, performing true single-pass softmax-averaged ensembling, moving the final resampling to the GPU, and evaluating TensorRT or ONNX Runtime export.

The author intends to register at the opening of the STSR 2027 registration window and would welcome feedback on the methodology and the analysis presented here.

Data Use and Reproducibility

The dataset was obtained through the challenge's official access procedure, under the signed Data Access Agreement, and was used solely for the purposes of this challenge participation. No portion of the data was redistributed, published, or used for any other project. All pipeline scripts, the training configuration, and the cross-validation results reported here are available from the author on request.

References

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Appendix A — Per-case cross-validation results

Each case was predicted exclusively by the fold that excluded it from training. Dice is averaged over classes present in that case's ground truth.

Case	Fold	Mean Dice	Teeth	Pulp	Classes
case_0000	4	0.630	0.748	0.394	45
case_0001	2	0.661	0.732	0.490	41
case_0002	2	0.636	0.770	0.368	48
case_0003	0	0.427	0.489	0.302	48
case_0004	3	0.635	0.837	0.231	42
case_0005	4	0.526	0.702	0.328	53
case_0006	3	0.655	0.768	0.302	37

Case	Fold	Mean Dice	Teeth	Pulp	Classes
case_0007	1	0.751	0.860	0.538	44
case_0008	3	0.425	0.456	0.383	56
case_0009	1	0.751	0.883	0.623	61
case_0010	0	0.740	0.886	0.584	60
case_0011	0	0.770	0.878	0.629	51
case_0012	4	0.557	0.718	0.311	53
case_0013	3	0.605	0.831	0.335	55
case_0014	1	0.730	0.726	0.735	59
case_0015	2	0.378	0.326	0.436	53
case_0016	0	0.730	0.841	0.606	51
case_0017	2	0.777	0.863	0.666	57
case_0018	0	0.398	0.342	0.459	62
case_0019	0	0.779	0.920	0.632	61
case_0020	0	0.845	0.894	0.790	51
case_0021	3	0.495	0.780	0.211	44
case_0022	0	0.556	0.627	0.485	22
case_0023	4	0.501	0.655	0.347	14
case_0024	2	0.775	0.853	0.697	18
case_0025	2	0.623	0.605	0.654	38
case_0026	2	0.469	0.612	0.293	49
case_0027	0	0.523	0.590	0.453	47
case_0028	3	0.560	0.838	0.261	54
case_0029	4	0.678	0.705	0.647	21

Table A1. Fold-aware cross-validation, all 30 annotated cases.